Hybrid Blockchain-Based Voting System with Machine Learning-Based Fraud Detection.

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Abstract

Traditional voting methods and many existing electronic voting systems face significant challenges such as identity fraud, duplicate voting, vote tampering, delayed result processing, and lack of transparency, which undermine public trust in digital elections. This paper proposes a hybrid e-voting framework that integrates machine learning, biometric authentication, and blockchain technology to provide a secure, transparent, and reliable solution. The system employs a Convolutional Neural Network (CNN) based facial recognition model as a biometric gatekeeper to authenticate voters by calculating a confidence probability, ensuring robust identity verification and minimizing false alarms through costsensitive learning. Suspicious activities like low-confidence matches, multiple voting attempts, and unauthorized access like trigger security alerts, while a validation layer prevents duplicate voting. Once authenticated, votes are securely stored on a blockchain, where each vote is hashed and linked to previous blocks to ensure immutability and detect tampering. The system continuously monitors the hash chain integrity, sending warnings if any unauthorized modifications occur. Additional security measures such as rate limiting and session locking further protect against bot attacks. This framework demonstrates a practical application of machine learning and blockchain in e-voting, addressing critiques on AI usage and blockchain effectiveness by emphasizing biometric probability-based authentication and blockchain's essential role in vote integrity.

Keywords: *Hybrid E-voting, Facial Recognition, Machine Learning, Blockchain Security, Biometric Authentication*

Introduction

Blockchain-based voting systems have emerged as a compelling alternative to conventional electoral mechanisms by recording each vote in an immutable and verifiable ledger. This approach reduces risks of tampering, duplication, and fraud while enhancing transparency throughout the voting process. The decentralized nature of blockchain, reinforced by cryptographic validation, ensures that no single authority can alter voting records, thereby significantly increasing trust and reliability in digital elections. As highlighted in [1], these

Overall, the system maintains a balance between security and usability. While achieving high accuracy, it avoids overfitting to ensure reliable real-world performance under variations such as lighting or facial changes. The integration of SQLite with blockchain also improves efficiency, enabling fast processing for registration and voting. The results confirm that the system is suitable for secure, scalable, and transparent small-to-medium scale electronic voting applications.

Conclusion

The proposed system successfully integrates CNN based facial recognition, liveness detection, and blockchain technology to ensure secure and tamper-proof electronic voting. It achieves high biometric accuracy of 93% with string fraud prevention, including false acceptance and effective liveness verification. The blockchain layer guarantees vote immutability through SHA-256 hashing and detects any tampering instantly. The system is limited by moderate CNN accuracy under extreme lighting or facial changes and is best suited for small-to-medium scale deployments rather than large national elections. Overall, the system provides a reliable balance of security, efficiency, and usability, making it suitable for digital voting applications with transparent and verifiable results.

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